

ANR ROBINSON INTL, Trinidad and Tobago

WMO: 789613

Lat: 11.150N

Lon: 60.833W

Elev: 40

StdP: 14.67

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00410

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	73.0	73.4	64.5	91.1	77.8	65.9	95.6	77.8	24.3	84.2	22.7	83.8	3.1	60	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	9.5	89.6	79.1	88.6	78.7	88.1	78.5	80.7	87.2	80.1	86.6	79.8	86.3	13.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.0	150.5	85.5	78.4	147.7	85.0	77.4	142.7	84.0	44.4	87.2	43.8	87.2	43.3	87.0	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.4	19.6	18.3	69.2	91.6	3.3	1.2	66.8	92.4	64.9	93.1	63.0	93.7	60.6	94.6		
(5)			65.8	82.2	2.4	1.0	64.0	82.9	62.6	83.5	61.2	84.1	59.5	84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	79.7	79.7	80.6	81.7	82.5	81.9	81.9	82.4	83.0	82.4	81.6	80.5	(6)
(7)		DBStd	1.85	1.27	1.44	1.58	1.46	1.41	1.59	1.45	1.59	1.59	1.64	1.60	1.39	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11505	921	832	948	952	1009	958	990	1005	991	1005	947	947	(10)
(11)		CDD65	6030	456	412	483	502	544	508	525	540	541	540	497	482	(11)
(12)		CDH74	61303	3746	3385	4430	5179	5969	5441	5584	6032	6265	5946	4984	4342	(12)
(13)		CDH80	18081	883	802	1161	1521	1878	1565	1643	1987	2238	1932	1412	1059	(13)
(14)	Wind	WSAvg	8.5	8.0	9.0	9.6	10.5	11.1	10.2	7.9	6.6	7.3	7.3	7.1	7.3	(14)
(15)	Precipitation	PrecAvg	57.2	1.8	1.7	1.8	1.5	2.3	5.6	7.0	6.2	6.5	8.6	8.6	5.9	(15)
(16)		PrecMax	69.4	5.6	4.5	5.5	3.5	7.6	11.3	13.4	9.8	10.8	15.2	19.5	12.4	(16)
(17)		PrecMin	32.9	0.3	0.1	0.2	0.2	0.4	1.6	3.0	2.4	2.1	4.0	3.7	1.3	(17)
(18)		PrecStd	9.1	1.4	1.1	1.5	1.1	1.7	2.8	2.8	2.1	2.8	3.5	3.8	3.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.4	87.6	88.1	88.5	89.8	89.3	88.9	90.0	90.0	89.9	88.9	88.1	(19)
(20)			MCWB	76.5	76.4	76.4	77.5	78.0	78.2	78.4	79.3	79.6	79.6	79.2	78.2	(20)
(21)		2%	DB	86.2	86.2	87.2	87.9	88.2	88.0	88.0	89.1	89.5	88.6	88.0	86.8	(21)
(22)			MCWB	76.0	75.7	76.1	77.3	77.7	78.1	78.4	79.1	79.5	79.3	79.1	77.7	(22)
(23)		5%	DB	85.7	85.6	86.2	87.0	87.8	87.4	87.4	88.0	88.2	88.0	87.5	86.1	(23)
(24)			MCWB	75.7	75.4	75.6	76.8	77.6	77.9	78.3	78.9	79.0	79.2	78.9	77.3	(24)
(25)		10%	DB	84.4	84.4	85.6	86.1	86.4	86.2	86.2	87.4	87.8	87.5	86.2	84.9	(25)
(26)			MCWB	74.9	74.5	75.3	76.4	77.0	77.5	77.8	78.7	78.9	79.0	78.2	76.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	78.3	79.0	79.9	80.1	80.4	80.6	81.0	81.6	81.0	80.8	80.1	(27)
(28)			MCDWB	85.0	84.9	85.6	86.8	86.6	86.5	86.8	87.7	87.7	87.5	87.0	86.1	(28)
(29)		2%	WB	77.3	77.2	77.7	78.8	79.6	79.6	79.9	80.1	80.4	80.3	80.0	79.1	(29)
(30)			MCDWB	84.3	84.4	84.7	85.9	86.1	85.7	86.1	86.8	87.3	86.8	86.2	85.1	(30)
(31)		5%	WB	76.5	76.2	76.9	78.0	78.8	78.9	79.2	79.7	79.9	79.8	79.5	78.2	(31)
(32)			MCDWB	83.4	83.4	84.0	84.9	85.4	84.9	85.4	86.3	86.7	86.2	85.6	84.1	(32)
(33)		10%	WB	75.7	75.3	76.0	77.3	78.0	78.2	78.4	79.1	79.3	79.2	78.8	77.4	(33)
(34)			MCDWB	82.4	82.5	83.2	84.1	84.5	84.1	84.5	85.5	85.9	85.5	84.8	83.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.4	10.5	10.3	9.6	9.3	9.0	9.4	9.6	9.5	9.6	9.7	9.9	(35)
(36)			MCDBR	11.3	11.3	10.8	10.2	10.1	9.9	10.1	10.4	10.1	10.3	10.7	10.7	(36)
(37)			MCWBR	5.4	5.3	5.0	4.7	4.6	4.7	5.1	5.1	5.0	5.1	5.4	5.4	(37)
(38)		5% WB	MCDBR	10.4	10.6	10.2	9.5	9.2	9.1	9.6	9.9	9.8	9.9	10.1	10.0	(38)
(39)			MCWBR	5.6	5.6	5.3	4.9	4.8	5.0	5.3	5.4	5.2	5.3	5.5	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.383	0.405	0.400	0.448	0.470	0.524	0.491	0.439	0.417	0.392	0.383	0.395	(40)	
(41)		taud	2.578	2.486	2.518	2.382	2.311	2.129	2.248	2.455	2.512	2.612	2.630	2.567	(41)	
(42)		Ebn at Noon	293	290	292	275	264	246	256	273	282	290	291	286	(42)	
(43)		Edh at Noon	31	36	35	40	42	49	44	36	35	31	30	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1683	1888	1992	2036	1932	1800	1864	1918	1920	1731	1571	1550	(44)	
(45)		RadStd	86	80	94	81	126	89	106	83	88	78	109	93	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.36	+0.59	+0.74	+0.25	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page